Phase 1 Proposal: Lightweight Spike-Based DOA Sector Inference for Embedded Edge AI

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EE 6900 Embedded AI

July 2, 2026

Abstract

This proposal defines Phase 1 of a staged embedded-AI research and engineering project. The objective is to build a compact, reproducible Python reference prototype for direction-of-arrival (DOA) sector classification using a physics-based uniform linear array (ULA) model, beamspace feature extraction, rate-based spike encoding, and a lightweight leaky-integrate-and-fire (LIF)-inspired inference mechanism. The goal is not to claim a novel DOA estimator in Phase 1. Instead, this phase establishes a defensible baseline that connects array signal processing, sparse temporal encoding, and future embedded deployment on PYNQ or FPGA-oriented platforms.

1 Motivation

Embedded AI systems must often make decisions under limited compute, memory, energy, and latency budgets. Classical deep-learning models may provide high accuracy, but they are not always appropriate for edge sensing platforms where deterministic operation, reduced arithmetic complexity, and hardware-friendly inference are important. Spike-based or SNN-inspired inference provides a useful design direction because information can be represented temporally and sparsely.

The selected sensing case for Phase 1 is DOA sector classification. This choice is technically meaningful because it is grounded in array signal processing while remaining small enough for a focused Python prototype. A ULA-based DOA model also connects naturally to future radar, MIMO sensing, beamspace processing, and FPGA acceleration work.

2 Problem Statement

The Phase 1 problem is to classify the angular sector of a single narrowband source impinging on a uniform linear array. The prototype must compare a conventional non-spiking baseline against a sparse spike-encoded LIF-inspired classifier. The main engineering question is:

> Can compact beamspace features extracted from a physics-based ULA signal model be converted into sparse spike trains and classified with acceptable accuracy and controllable temporal latency?

3 Objectives

The Phase 1 objectives are:

1. Implement a reproducible ULA signal model for single-source DOA-sector classification.
2. Extract compact beamspace features from sample covariance matrices.
3. Train and evaluate a deterministic nearest-centroid non-spiking baseline.
4. Encode normalized features into sparse spike trains using rate-based encoding.
5. Implement a lightweight LIF-inspired classifier as a temporal evidence accumulator.
6. Evaluate accuracy versus SNR and estimate stable-decision latency in time steps.
7. Sweep LIF time-window and leak parameters to study the accuracy–latency trade-off.
8. Produce report-ready figures, tables, and a clean project handoff.

4 Technical Approach

4.1 Signal Model

The dataset will be generated from a narrowband single-source ULA model. For a source at angle θ, the array snapshot matrix is modeled as

$$X = a(\theta)s + N, \tag{1}$$

where $a(\theta)$ is the ULA steering vector, s is the narrowband source waveform, and N is circular complex Gaussian noise. The sample covariance matrix is

$$R_{xx} = \frac{1}{M} X X^H, \tag{2}$$

where M is the number of snapshots.

4.2 Feature Extraction

Beamspace energy features will be computed over a fixed angular grid. These features provide a compact representation of spatial information without implementing a full high-resolution DOA estimator in Phase 1.

4.3 Baseline Classifier

A nearest-centroid classifier will be used as the deterministic non-spiking baseline. This baseline is simple, reproducible, and interpretable. Its role is to determine whether the compact beamspace features contain enough information for coarse DOA-sector classification before spike encoding.

4.4 Spike Encoding and LIF-Inspired Inference

Normalized beamspace features will be converted into binary spike trains using rate-based encoding. The LIF-inspired classifier will accumulate spike-driven evidence in a leaky membrane state for each class. The predicted class is determined by the final membrane value, while stable-decision latency is estimated by identifying the earliest time step after which the final decision remains unchanged.

Table 1: Planned Phase 1 Dataset Configuration

Parameter	Value
Array type	Uniform Linear Array
Number of elements	$N = 8$
Snapshots per sample	$M = 128$
Element spacing	$d = \lambda/2$
Source count	1
DOA range	-60° to $+60^\circ$
Number of sectors	4
SNR values	$-10, -5, 0, 5, 10, 15, 20$ dB
Samples per sector per SNR	100
Total samples	2800
Train/test split	70/30, stratified

5 Initial Dataset Configuration

6 Evaluation Metrics

Phase 1 will use the following metrics:

- overall test accuracy,
- accuracy versus SNR,
- confusion matrix at a representative SNR,
- spike-count statistics,
- mean stable-decision latency in time steps,
- accuracy–latency trade-off under LIF parameter sweeps.

7 Expected Deliverables

The expected deliverables are:

1. Python source code for the end-to-end Phase 1 pipeline.
2. CSV and text result files for reproducibility.
3. Report-ready figures for accuracy, latency, and trade-off analysis.
4. A formal Phase 1 report.
5. A compact IEEE-style paper draft.
6. A presentation slide deck and transcript.

8 Expected Contribution

The expected contribution of Phase 1 is a clean engineering bridge between classical array processing and hardware-aware spike-based inference. The output should be suitable as a course deliverable and as a foundation for later phases involving richer radar/DOA case studies, PYNQ deployment, and Vitis HLS kernel exploration.

9 References

References

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